Program–Data Map (Replication Manifest)

> Management Science (INFORMS) submission — Code and Data Disclosure package README. 2026-08-11. This map lists all 40 scripts in `03_程序/` (organized into five purpose folders) and their corresponding data in `02_数据/`. It is the master index for replication.

Environment and running conventions

- **Python 3.13** (theory scripts are pure standard library; empirical and figure scripts require `numpy`; figure scripts require `matplotlib`). See `requirements.txt` for version-pinned environment (verified: numpy 2.5.1, matplotlib 3.11.1).
- Every script contains a directory bootstrap (`os.chdir(os.path.dirname(os.path.abspath(__file__)))`): **run from anywhere inside the package** — all data paths are relative (`../../02_数据/<folder>/...`); no machine-specific absolute paths remain.
- Data are organized by appendix: `02_数据/D3`, `D4`, `D6`, `D7`, `D8`, `arXiv`, `其他`.
- D.7 requires the six Amazon metadata files (5.9 GB, not shipped); download instructions and checksum anchors are in `02_数据/其他/数据来源说明.md` and `02_数据/D7/D7_数据说明.md`.
- All Chinese comments were removed from scripts at packaging (2026-08-11); console output of a few scripts remains English/Chinese-neutral.

1. Theory verification (01_theory/, 7 scripts, pure stdlib)

Script	Corresponds to	Data	Verified outputs
`audit_recheck.py`	Appendix B Tables B.3/B.4, full-chain audit, MP section (cited 4× in the paper)	none (built-in parameters)	K_μ*=3.9456, K̂=5.804, θ_s peaks 6.83/4.96/4.26, P3′ single-crossing
`muB_internalize.py`	Appendix B (endogenous μ_B; cited 3×)	none	ΔV single-crossing, unimodal μ_B, W valley 1.838964
`thetas_check.py`	§2.3 ρ_vol/ρ_val Hill saturation (cited 2×)	none	order-parameter arm peaks θ_s
`p3prime_sweep.py`	Appendix B Table B.4 (P3′ 486-parameter grid)	none	97.7% (475/486) feasible domain
`w_internalize.py`	Appendix B (platform-side w(K) endogenized)	none	suppression intensity w(K) from the platform FOC
`welfare_check.py`	§5 welfare valley	none	K_worst≈4.5403, subsidy-sign flip breakpoints
`formD_check.py`	Appendix A.2 (formal invariance Fit^D)	none	κ_A*=0.5418/0.3191/0.3743/0.2004/0.5314

2. Empirical D3/D6 (02_empirical_D3D6/, 10 scripts)

Script	Corresponds to	Data	Verified outputs
`d3_build_panel.py`	Table D.1 panel construction	writes `D3/d3_panel.json`	**WIPO-anchored** K_stock (140→1,109 thousand; unified metric)
`d3_regression.py` / `d3_regression2.py`	Table D.2 (D.3 regressions S1–S5)	reads `D3/d3_panel.json`	β1 differentiated response; τ_T interaction; PDD-robustness
`panel_extend.py`	D.6 test ① (8-platform take rates)	built-in	−9.45+3.92·lnK, R^2=0.980; +0.90/+0.35/+2.52 pp/yr
`d6_upgrade.py`	D.6 formal estimates (i)(ii)(iii)	reads `D3/d3_panel.json` → writes `D6/d6_upgrade_results.json`	Welch t=4.65/4.68 CI[13.8,35.4]; panel β=+1.0753 p_wild=0.150; C-segment +3.9238 R^2=0.9804
`amazon_break.py`	D.6 test ③ (quarterly break)	built-in → writes `D6/amazon_break_results.json`	YoY 43.0→20.8%; Welch t=2.37; Chow F(6,6)=7.96
`amazon_break_seasonal.py`	same (seasonally adjusted reproduction)	built-in	43.0→20.8% / t=2.37 / Chow 7.96
`d3_variants.py`	D.6 (v-a)–(v-d) mechanism variants	reads `D3/d3_panel.json` → writes `D3/d3_variants_results.json`	v-b β1=+0.31; v-c p>0.7; v-d +0.61/+0.06/+0.07/−0.46/−0.24
`d3_variants_v6.py`	D.6 (v-a) contamination-label interaction (independent reproduction)	reads `D3/d3_panel.json`	**β2=−0.70 (p_wild=0.062)**; lag2 −0.71 (0.063)
`e3_regression.py`	D.6′ (DiD diagnosis)	built-in	R^2=1.000 numerical artifact reproduced; PDD-robustness fails

3. Empirical D7/D8 (03_empirical_D7D8/, 3 scripts)

Script	Corresponds to	Data	Verified outputs
`shelf_final.py`	D.7 six-category shelf test (sole numerical source)	reads `D7/` (download 6 meta files)	six slopes −2.753/−4.424/−3.674/−0.992/−1.881/−1.014; 59.3% excluded; 4/4 monotone; Test3
`repair_check.py`	D.8 ①③④ (repair-domain pilot)	reads `D8/heimao_sample.csv`, `D8/court_evidence.csv`	① t=1.01; ③ monotone 0%/95%/100%; ④ 40.5/29.7/16.2/13.5
`repair_sens.py`	D.8 ② (mechanism-consistent subsample)	reads `D8/heimao_sample.csv`	+0.0815 (t=2.57)/+0.2551 (t=9.29)/+0.2371 (t=9.13)

4. Figures (04_figures/, 7 scripts; outputs → `01_投稿材料/Figures_EN/`)

> Note: `figures_en.py` regenerates all of Fig1–Fig5; Fig1_theta_U is a historical exhibit not referenced in the paper (the submission contains 8 figures). If a run produces an extra `Fig1_theta_U.png/svg`, ignore

> or delete it.

Script	Output	Corresponds to
`fig2_rebuild.py`	Fig2_welfare_valley.png/svg (current version)	Figure 2 (§5.2)
`figures_en.py`	Fig3_projection_tower, Fig5_dual_timescale_welfare (plus original Fig1/2/4)	Figure 3 (Cor. 1.1)/Figure 5
`fig4_rebuild.py`	Fig4_observational_misalignment.png/svg (current version)	Figure 4 (§2.2)
`fig_llm_break.py`	FigD6_llm_break.png/svg	Figure D.6.1 (LLM break)
`fig_suppression_prices.py`	FigD6_suppression_prices.png/svg	Figure D.6.2 (suppression prices)
`fig_industry_tau.py`	FigD6_industry_tau.png	Figure D.6.3 (OpenAlex industry cross-section)
`fig_arxiv_tau_panel.py`	FigD6_arxiv_tau_panel.png	Figure D.6.4 (arXiv cross-industry diffusion + Poisson)

5. arXiv / τ_T / OpenAlex (05_arxiv_tauT/, 13 scripts)

Script	Corresponds to	Data	Verified outputs
`fetch_arxiv.py` / `fetch_arxiv2.py`	§4.2/Appendix D τ_T proxy	online API → writes `arXiv/arxiv_counts.json`	arXiv category counts (2014–2024)
`tau_analysis.py`	§4.2/Appendix B τ_T empirical proxy	reads `arXiv/arxiv_counts.json`	τ_T series; recognition share 18.4→14.9→19.0%
`five_paths.py`	§6 evidence-chain five paths	reads `arXiv/arxiv_counts.json`	five-path synthesis
`arxiv_industry_panel.py`	D.6.4 industry cross-section	writes `arXiv/arxiv_industry_panel.json`	industry-level τ_T panel
`arxiv_panel_analysis.py`	D.6.4 panel analysis	reads `arXiv/arxiv_industry_panel.json` → writes `arXiv/arxiv_panel_analysis.json`	Δ +18.2 (p=0.039)/+10.8 (0.070)/+3.9 (0.064); both p>0.7
`arxiv_poisson.py`	D.6.4 Poisson diffusion	reads `arXiv/arxiv_panel_analysis.json`	β(post×sparse)=+0.203 (t=2.06, p=0.039)
`arxiv_robust.py` / `arxiv_robust_summary.py`	D.6.4 placebo/sensitivity	reads `arXiv/arxiv_industry_panel.json`	generation-class placebo +10~+28pp; category sensitivity +15~+24pp
`d4_cross_reg.py`	L259 OpenAlex cross regression	writes `D4/d4_cross_reg.json`	ρ=0.33 (p=0.25); β=−0.16 (t=−0.32)
`d4_industry_tau.py` / `d4_industry_ext.py`	L259 industry cross-section (edu 10.9%/recruiting 11.6%/e-commerce 45.3%)	writes `D4/d4_industry_tau.json`, `D4/d4_industry_tau_ext.json`	industry τ_T and recognition shares

6. Data files by folder (02_数据/)

Folder	Files	Generated by
`D3/`	d3_panel.json, d3_results_jd.json, d3_variants_results.json	d3_build_panel / d3_regression* / d3_variants
`D4/`	d4_cross_reg.json, d4_industry_tau.json, d4_industry_tau_ext.json	d4_cross_reg / d4_industry_tau / d4_industry_ext
`D6/`	d6_upgrade_results.json, amazon_break_results.json	d6_upgrade / amazon_break
`D7/`	D7_数据说明.md (raw meta files to be downloaded)	read by shelf_final
`D8/`	heimao_sample.csv, court_evidence.csv	read by repair_check / repair_sens (hand-coded samples; alternative disclosure plan)
`arXiv/`	arxiv_counts.json, arxiv_industry_panel.json, arxiv_industry_tau.json, arxiv_panel_analysis.json, arxiv_robust.json	fetch_arxiv / *arxiv_*
`其他/`	1.txt, 数据来源说明.md (data-source note)	—

7. Recommended replication order

```
01_theory/*.py                           # pure stdlib; run directly
02_empirical_D3D6/d3_build_panel.py → d3_regression*.py → d6_upgrade.py → d3_variants.py → d3_variants_
02_empirical_D3D6/amazon_break*.py, panel_extend.py, e3_regression.py
03_empirical_D7D8/repair_check.py, repair_sens.py   # data shipped in 02_数据/D8
03_empirical_D7D8/shelf_final.py                   # download 6 meta files to 02_数据/D7 first
04_figures/*.py                                    # outputs to 01_投稿材料/Figures_EN/ (numpy+matplot
05_arxiv_tauT/*.py                                 # fetch_arxiv* need network; others read shipped J
```

8. Disclosure notes

- **Reproducible material**: 40 scripts + constructed data (JSON/CSV) shipped in this package; the map above links every main-text result to its generating script and data.
- **Third-party public data** (not shipped, identified in the manuscript): Amazon Review Data (2018) metadata (download links and checksum anchors in `数据来源说明.md`), platform 20-F/annual-report filings, WIPO and arXiv aggregates.
- **Alternative disclosure plan (proprietary/semi-public)**: the D.8 repair-domain samples (`D8/heimao_sample.csv`, n=98; `D8/court_evidence.csv`, n=37) are hand-collected/coded from public complaint and court platforms; samples and the coding scheme are shipped, but fully automated re-collection is not feasible — flagged for the Data and Code Disclosure form as an exception request with the samples attached.
- Historical/development scripts (blk*, p1_*, p2_main, shelf_ *exploration chain, md2docx_*, etc., 33 files) are archived in `05_历史版本链/开发脚本_20260811/` and are not part of the replication set.